Fibre Assurance & Network Operations

Solution Overview & Commercial Value Proposition

Product Brief

Overview

Carrier NOC teams spend too much time stitching together separate screens across optical NMS, IP routers, GIS maps, OTDR testers, and ticketing tools. When an alarm storm hits, it routinely takes 30 to 60 minutes just to tell whether a fibre trunk was actually cut, which enterprise customers are affected, and where the field crew needs to dig.

Our platform connects these existing data sources into one operational view. It links customer circuits directly to physical ducts, filters out false dispatches caused by power or maintenance events, gives splice teams accurate street coordinates, and helps pre-sales teams verify route diversity in seconds.

Where Carrier Operations Lose Time and Money Today

- **No warning before a cut:** NOCs only find out about roadwork when a backhoe snaps the cable, turning a preventable threat into a multi-hour civil repair.
- **Inaccurate dig locations:** Raw OTDR distance measures coiled glass, not street distance. Ignoring 15 to 25 meters of slack in each manhole regularly sends crews 150 to 300 meters away from the break.
- **Wasted truck rolls:** Building power blackouts, heavy rain on microwave links, and overrun maintenance windows trigger alarms that look like fibre faults, wasting field dispatches.
- **Hidden single points of failure:** Primary and backup circuits sold as "diverse" often end up sharing a last-mile chamber, bridge crossing, or landing corridor.

Six Practical Capabilities

1. **Early warning before a fibre cut.** Watches live optical polarization shifts and vibration patterns along buried ducts. When heavy digging starts near a trunk, the NOC gets an early alert to shift traffic onto a backup path before the cable breaks.

2. **Street-accurate fault pinpointing & false-alarm filtering.** Reads OTDR traces against actual GIS splice records, subtracting manhole slack loops to give crews exact GPS coordinates. Automatically separates real physical breaks from customer power outages and weather fade.

3. **Instant route-diversity checks for Pre-Sales.** Lets sales engineers enter two sites on a map and confirm in seconds that primary and backup paths share zero manholes, ducts, or bridges, exporting a clean route and latency summary for customer RFPs.

4. **Subsea coastal cable & anchor watch.** Overlays live ship tracking (AIS) onto submarine landing corridors alongside cable power and optical health, alerting operators when a vessel slows down or drags anchor near shallow-water segments.

5. **Maintenance & roadwork cross-check.** Checks incoming alarms against active change tickets and municipal roadwork permits within 500 meters so engineers immediately see when an outage is tied to planned work or street excavation.

6. **Field crew handoff, splice sign-off & SLA reports.** Sends the technician the exact cable, tube/core colors, and realistic repair window, then runs a remote OTDR test to verify splice loss before closing the ticket. Drafts incident reports (English, Arabic, or wholesale format) and flags outages exempt from SLA credits.

How Value Maps to Different Operators

Operator Profile	Typical Operational Pain	Where the Platform Helps Most
Metro & Dark Fibre (e.g., Constl)	Frequent urban JCB cuts; strict latency SLAs for banks and data centers; slow manual GIS checks during RFPs.	Pre-cut digging alerts, street-accurate splice dispatch, and instant zero-shared-duct route sheets for enterprise bids.
National Converged Carriers (e.g., e&, du, Vi)	Thousands of mixed optical, IP, and 5G alarms during faults; wasted dispatches on site power loss; SLA reporting load.	Collapses alarm noise into one root cause, separates power/weather events from fibre cuts, and automates EN/Arabic RFOs.
Subsea & Wholesale (e.g., PCCW Global)	Shallow-water anchor damage; slow handoffs between subsea landing stations and terrestrial backhaul teams.	Live vessel tracking over cable corridors, clear wet-plant vs. land fault isolation, and standard wholesale RFO packs.

Expected Operational & Commercial Impact

Fewer wasted truck rolls

Keep fibre splice vans from rolling out on customer building power failures, microwave rain squalls, or active maintenance windows.

Faster repair times (MTTR)

Cut NOC triage from 45 minutes to a couple of minutes, and send crews to the right chamber on the first visit rather than walking the street.

Faster enterprise quote turnaround

Give pre-sales teams verified diverse routes and optical latency numbers in minutes instead of waiting days for manual GIS reviews.

Protected SLA margins & clean repairs

Catch high-loss contractor splices before crews leave the trench, and document third-party or customer-power faults to avoid unnecessary SLA credits.

Integration & Operating Principles

- **Works with existing network tools:** Connects to standard optical controllers (Ciena, Nokia, Huawei), OTDR systems (VIAVI, EXFO), GIS inventory exports, IP syslog/SNMP feeds, and ITSM platforms (ServiceNow, Jira).
- **Grounded in network data, not guesswork:** Fault locations, slack calculations, and backup routes are computed directly from GIS geometry, OTDR traces, and network topology. Text generation is only used at the final step to summarize verified findings and draft customer reports.
- **Human engineer stays in control:** The software prepares traffic switchover commands, field dispatches, and incident reports, while NOC staff review and approve every action before execution.

Suggested 4-Week Pilot

The simplest way to evaluate value is a read-only trial on a single metro ring or inter-city corridor:

- **Weeks 1–2:** Load GIS duct/cable records for one corridor and connect read-only alarm and OTDR feeds.
- **Week 3:** Run live alongside the NOC to benchmark fault location accuracy, false-dispatch filtering, and pre-sales diversity checks.
- **Week 4:** Review measured time savings and dispatch accuracy with operations and engineering leadership.